

Mixture of experts framework for domain adaptive colorization

Project Proposal

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Abstract

A vast majority of human history has been captured exclusively in black and white, through the lenses of early photo/video capturing devices, leaving generations of visual records stripped of the chromatic context that shapes how we perceive and connect with the past. While modern deep learning has made grayscale-to-color conversion feasible, training a single model to pseudocolor the infinite diversity of everyday life forces the network to approximate an impossibly broad color manifold, ultimately collapsing into a mediocre generalist that fails every specialist domain it attempts to serve. This project proposes a Mixture-of-Experts (MoE) architecture that reframes pseudocoloring not as a monolithic prediction task, but as a divide-and-conquer problem. By distributing the burden of colorization across specialized experts rather than burdening a single network with the entirety of human visual experience, this approach aims to recover historically authentic and perceptually convincing color from our monochrome past with a fidelity that generalist models fundamentally cannot achieve.

1. Introduction

Image colorization restores plausible color to grayscale photographs, serving applications in historical preservation, digital media, and visual analytics. While deep learning has produced increasingly photorealistic results, the task remains fundamentally challenging because the mapping from luminance to chrominance is highly underconstrained and varies dramatically across visual domains.

Portraits/people, landscapes, urban scenes and animals each possess distinct color distributions and semantic structures that place conflicting demands on a unified model. Recent work has emphasized the importance of global context, local precision, and multimodal color priors for high-quality results. Mixture-of-Experts architectures offer a compelling alternative by partitioning complex problems across specialized sub-networks, though their application to colorization remains largely unexplored.

This project investigates a Mixture of Experts framework for general-purpose image colorization. By decomposing images into semantically coherent regions and routing them to domain-specialized experts, we seek to improve chromatic fidelity and semantic consistency across diverse scenes. The framework will explore strategies for spatial decomposition, expert specialization, and coherent output integration, initially scoped to photographic colorization with future extension to video and medical imaging.

2. Goals and Objectives

This project aims to develop a Mixture of Experts framework that converts grayscale photographs into realistic color images by leveraging domain-specialized expertise rather than relying on monolithic prediction networks.

Objectives:

1. To investigate effective strategies for identifying and isolating semantically distinct regions within grayscale images that belong to four colorization domains: portraits and people, nature and landscapes, urban and man-made scenes, and animals, measured by segmentation mIoU on a labeled dataset such as ADE20K or COCO.

2. To develop and train a specialized colorization model for each of the domains identified in Objective 1, so that every region isolated from an image is colorized by an expert trained on that domain, measured by FID, LPIPS, colorfulness, and PSNR/SSIM or Lab color difference (ΔE) against ground truth on held-out test images from each domain.
3. To design a mechanism that assigns image regions to the most suitable expert based on their visual and semantic characteristics, measured by routing accuracy (the fraction of regions assigned to the correct expert) on labeled test images.
4. To implement a strategy for combining expert outputs into a single color image that preserves smooth transitions, global color harmony, and overall photorealism, measured by color discontinuity (ΔE) across region boundaries, and by overall FID and LPIPS of the final image with and without the blending module.
5. To architect the framework so that new domain experts can be incorporated without requiring a complete redesign of the underlying system, demonstrated by adding a new expert and its prompt set without retraining or modifying the existing experts.

3. Scope of the Project

The project covers building a working colorization system for grayscale photographs, primarily across four image types: portraits/people, nature/landscapes, and urban /man-made scenes and animals. Rather than treating these as separate model categories, the system decomposes each image into semantic regions and colorizes each region with a domain-specialized expert, so that a single photograph can draw on several experts at once.

The work includes collecting training data for each expert domain (portraits and people, nature and landscapes, urban and man-made scenes, and animals), along with a segmentation-based routing module that assigns each region to the most relevant expert. Separate colorization models will be trained for each domain. Each expert receives its assigned region together with an added boundary of approximately 20% around it, so that local context is preserved at region edges. A blending module then combines the expert outputs into a single, consistent color image.

The project will produce a working prototype that demonstrates this multi-expert approach to colorization. The system will be designed so that new domain experts can be added later without rebuilding the entire system. Evaluation criteria for each objective are specified in Section 2.

4. Initial Study and Work Done so Far

Image colorization has seen rapid progress across convolutional, transformer, generative adversarial, and diffusion-based architectures, with the field now encompassing automatic, exemplar-guided, text-conditioned, and multimodal paradigms [1].

Contemporary non-MoE research has established several principles for high-quality colorization. The mapping from grayscale to color is inherently multimodal, and explicitly modeling multiple plausible color hypotheses produces more vibrant results than deterministic regression [2]. Coherent colorization also requires coupling global scene semantics with local spatial features, as demonstrated by dual-decoder architectures that separately handle semantic reasoning and pixel-level color restoration [3], while transformer-based approaches can capture the long-range dependencies essential for

context-aware prediction [4]. Additionally, generating vivid rather than desaturated outputs requires explicit priors that encourage chromatic diversity [5].

Despite these advances, most existing approaches rely on a single monolithic network. Such a model must approximate an overly broad and often conflicting color manifold across disparate visual domains, which tends to produce desaturated outputs, semantic inconsistencies, or a regression toward a dull chromatic mean.

This limitation has motivated increasing interest in Mixture-of-Experts architectures. Fan et al. [6] introduced RS-MoEGAN, which employs sparse token-level routing in a Vision Transformer to decouple conflicting generation objectives across different land-cover types. Zhou et al. [7] proposed Gamma, using shared and adaptive experts to handle domain-specific biases across diverse image assessment tasks through mixed-dataset training. He [8] further established that scaling to a large number of small experts via product-key retrieval improves parameter efficiency without proportional computational cost.

Based on these ideas, we want to try a different approach with MoE. Instead of routing tokens inside transformers or mixing dataset-level adapters, we want to break the image into regions and send each region to a domain-specific expert. This way each expert only needs to learn a small part of the colorization problem, and we can still combine everything into one final image.

Datasets. So far we have identified public datasets for four expert domains: portraits and humans (CelebA, CelebA-HQ, FFHQ for faces; DeepFashion-MultiModal and COCO person images for full-body), nature and landscapes (Places365, SUN397), urban and man-made scenes (Cityscapes, ADE20K, urban subsets of Places365), and animals (AFHQ, Oxford-IIIT Pets, animal subsets of COCO). Each expert is trained on whole color images from its own domain, and grayscale inputs are derived from these images through the L channel in Lab space, so no manual annotation is needed for expert training. A labeled segmentation set such as ADE20K or COCO will be used to evaluate the quality of the region decomposition.

Model architecture. Region decomposition is handled by Grounded-SAM [9], which combines open-vocabulary object detection with promptable segmentation. Predefined text prompts associated with each domain (for example person, animal, building, sky, vegetation) are built into the system, so no manual input is required from the user. These prompts generate region masks. Each mask is used to crop the corresponding region with an added 20% boundary, and the crop is routed to the most relevant expert. Because Grounded-SAM is used pretrained, no router training is required, and new experts can be added by defining new prompts. Each expert is a colorization network that colorizes the region assigned to it, drawing on the surrounding visual context so that colors stay coherent across region boundaries. A blending module then merges the expert outputs into a single image, using soft mask boundaries to avoid visible seams between regions. As the quality of the segmentation ultimately determines the quality of the colorization pipeline, the router remains the component we will examine most closely.

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